

Cathode-Ray Oscilloscope (Cambridge (CIE) A Level Physics): Revision Note



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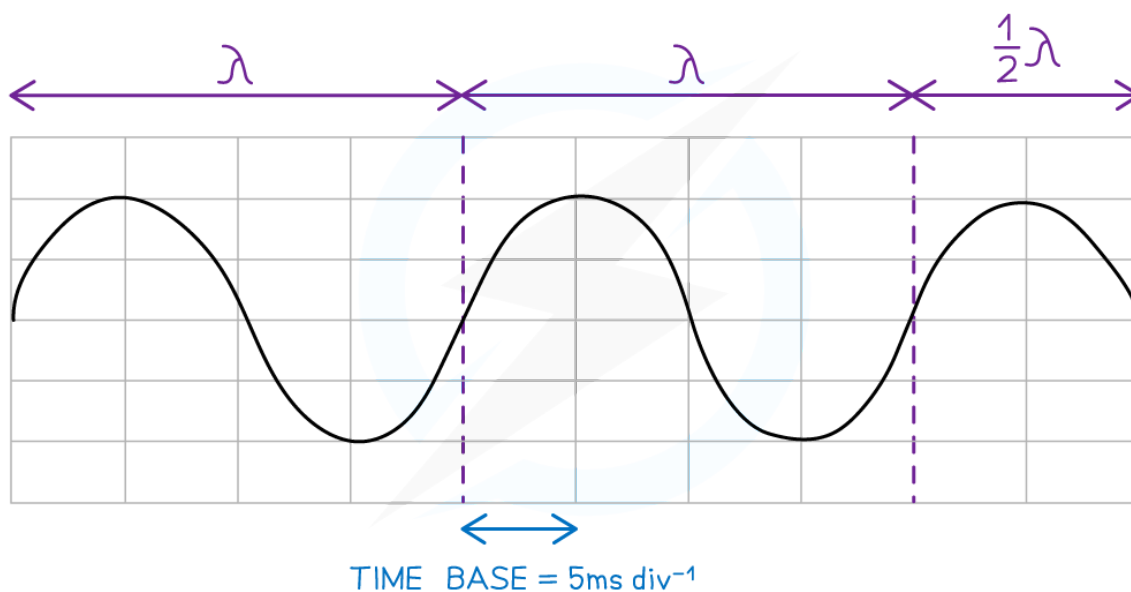
Reviewed by: Caroline Carroll

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Cathode-ray oscilloscope

- A cathode-ray oscilloscope is a laboratory instrument used to display, measure and analyse waveforms of electrical circuits
- An **a.c.** current on an oscilloscope is represented as a transverse wave
 - The oscilloscope can be used to determine its frequency and amplitude
- A **d.c** voltage on an oscilloscope is represented as a **horizontal line** at the relevant voltage
- The x-axis is the **time** and the y-axis is the **voltage** (or **y-gain**)

Cathode-ray oscilloscope display



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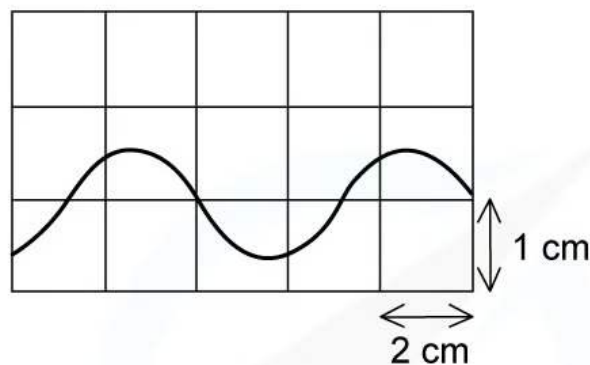
Diagram of cathode-ray oscilloscope display showing wavelength and time-base setting

- The period of the wave can be determined from the **time-base**
 - Time-base is **how many seconds each division represents** measured commonly in **s div⁻¹** or **s cm⁻¹**
- Count the number of wavelengths shown on the screen
- Divide the total time by the number of wavelengths to obtain the time period T
- Calculate the frequency of the wave using $f = \frac{1}{T}$



Worked Example

A cathode-ray oscilloscope (c.r.o.) is used to display the trace from a sound wave. The time-base is set at $7 \mu\text{s mm}^{-1}$.



What is the frequency of the sound wave?

- A. 2.4 Hz
- B. 24 Hz
- C. 2.4 kHz

D. 24 kHz

Answer: C

Step 1: Determine the period of the wave:

- One full wave covers three horizontal squares, which is a distance of 6 cm
- There are 60 mm in 6 cm
 - Each mm represents a time of 7 μs
 - Time period, $T = 60 \times (7 \times 10^{-6}) = 4.2 \times 10^{-4} \text{ s}$

Step 2: Use the frequency equation to determine frequency:

- Recall the frequency equation

$$f = \frac{1}{T} = \frac{1}{4.2 \times 10^{-4}} = 2380 \text{ Hz}$$

- Recall that 1000 Hz = 1 kHz and the answer must be given to 2 significant figures

$$f = 2.4 \text{ kHz}$$



Examiner Tips and Tricks

The time-base setting varies with units for seconds (commonly ms) and the unit length (commonly mm). Unit conversions are very important for the calculation of the time period and frequency